Lab 4 - Lunar Lander

AI-assisted work record | Vishnu Maddirala | SRN PES2UG24CS593 | 15 October 2026

Record type: Retrospective summary of the work actually performed in this chat. It is not a verbatim or reconstructed prompt-by-prompt transcript. The linked ChatGPT page contains the real conversation.

1. Assignment and repository

The allocation workbook maps SRN PES2UG24CS593 to repository 32_lunarLander. The repository README specifies a touchdown horizontal-speed fix plus three features: fuel-sensitive ship color, a successful-landing effect, and an extra-life score threshold.

2. Implementation

README item	Change made
Landing validation	Added abs(horizontal speed) <= MAX_SPEED_X to the safe touchdown conditions.
Fuel color	Interpolate the hull from pale blue at full fuel to red at empty fuel; clamp the fuel fraction to [0, 1].
Landing feedback	Create a score-colored particle burst at the landing position and draw it briefly after touchdown.
Bonus life	Return a 1,500-point threshold; existing score bookkeeping awards one life per newly crossed threshold.

3. Checks and evidence

Python syntax compilation succeeded. A behavior check confirmed that high sideways speed crashes, a safe landing succeeds, the fuel-color endpoints are correct, and crossing 1,500 points awards one life. Both MP4 files were confirmed as 10 seconds, 30 fps, 800 x 600.

Video method: The before and after clips are scripted demos rendered from the Pygame drawing code. They stage the target outcomes; they are not unedited screen recordings of a person playing.

4. Repository status

The Lab-4 files are committed on branch main as be68fe18eb7282cd799273eb7a47cf30104ec076, with origin set to the student's personal repository. The student subsequently confirmed that the push completed. The local main branch tracks origin/main at this commit.

Personal repository

https://github.com/vishnu-1107-cloud/SE-Lab4

ChatGPT history link supplied by student

https://chatgpt.com/share/6ad11b67-2bac-83ec-aa74-ea2aff62043c

Submission must follow the course instructions: place these deliverables under the personal repository's Lab-4 folder and do not open a pull request against SETAPESU26.